

Numerical Simulation of Quantized Vortices with Small Core Size via Vortex Tracking

Complex Quantum Systems and Vortex Dynamics
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1 Introduction

1.1 The Gross–Pitaevskii equation

Gross–Pitaevskii (GP) equations are nonlinear Schrödinger equations that arise in the modelling of quantum many-body systems, most prominently in Bose–Einstein condensation and superconductivity. A prototypical form is derived from the *Ginzburg–Landau energy*

$$E_\varepsilon(u) = \int_\Omega \frac{1}{2} |\nabla u|^2 + \frac{1}{4\varepsilon^2} (1 - |u|^2)^2 \, dx, \quad (1)$$

defined on a simply connected, smoothly bounded domain $\Omega \subset \mathbb{R}^2$. The parameter $\varepsilon > 0$ controls the *vortex core size*: smaller ε means sharper, more localised vortices.

The corresponding Schrödinger flow $i \partial_t \psi = -\nabla_{L^2} E_\varepsilon(\psi)$ on Ω with homogeneous Neumann boundary conditions reads

$$i \partial_t \psi_\varepsilon = \Delta \psi_\varepsilon(x, t) + \frac{1}{\varepsilon^2} (1 - |\psi_\varepsilon(x, t)|^2) \psi_\varepsilon(x, t), \quad x \in \Omega, \, t \geq 0. \quad (2)$$

This equation is globally well-posed for every $\varepsilon > 0$ and preserves the energy: $E_\varepsilon(\psi_\varepsilon(t)) = E_\varepsilon(\psi_\varepsilon(0))$ for all $t \geq 0$.

1.2 Physical quantities and quantized vortices

The key physical quantities associated with a wave function ψ are the *supercurrent*

$$j(\psi) = \frac{1}{2i} (\bar{\psi} \nabla \psi - \psi \nabla \bar{\psi})$$

and the *vorticity* (Jacobian)

$$J\psi = \frac{1}{2} \nabla \times j(\psi) = \det \nabla \psi.$$

A striking feature of GP dynamics is the appearance of *quantized vortices*: isolated zeros of ψ around which $j(\psi)$ winds with integer topological degree $d_j = \pm 1$. The density $|\psi|^2$ drops to zero in a small core of radius $\sim \varepsilon$ around each vortex location.

In the singular limit $\varepsilon \rightarrow 0$, the vorticity concentrates as

$$J\psi_\varepsilon(t) \rightarrow \pi \sum_{j=1}^N d_j \delta_{a_j(t)},$$

and the vortex positions $a(t) = (a_1(t), \dots, a_N(t))$ solve the *point vortex system*

$$\dot{a}_j(t) = -\frac{1}{\pi} d_j J_{\text{symp}} \nabla_{a_j} W(a(t), d), \quad a_j(0) = a_j^0, \quad (3)$$

where

$$J_{\text{symp}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is the symplectic matrix and W is the renormalized energy defined below. Equation (??) is a Hamiltonian ODE in $\mathbb{R}^{2N}$: J_{symp} rotates the gradient by 90° , so vortices move *perpendicular* to ∇W rather than along it, causing same-sign vortices to orbit each other.

1.3 The canonical harmonic map u^*

The limiting object of ψ_ε as $\varepsilon \rightarrow 0$ is the *canonical harmonic map*. It is not a modelling ansatz but the proven limit: $\psi_\varepsilon(t) \rightarrow u^*(a(t), d)$ in $W^{1,p}(\Omega)$ for $p < 2$ (Colliander–Jerrard, Lin–Xin). Given vortex positions $a \in \Omega^{*N}$ and degrees $d \in \{\pm 1\}^N$, we want a map $u^*: \Omega \rightarrow \mathbb{C}$ that

- satisfies $|u^*| = 1$ away from the vortex locations (i.e. lives on the unit circle S^1),
- has topological winding number d_j around each a_j , and
- satisfies the homogeneous Neumann condition $\partial_\nu u^* = 0$ on $\partial\Omega$.

These conditions uniquely determine the supercurrent $j(u^*)$ via

$$\nabla \cdot j(u^*) = 0, \quad (4)$$

$$\nabla \times j(u^*) = 2\pi \sum_{j=1}^N d_j \delta_{a_j}, \quad (5)$$

$$\nu \cdot j(u^*) = 0 \quad \text{on } \partial\Omega. \quad (6)$$

This is precisely the 2D incompressible Euler vorticity equation with point vortices. Via the Biot–Savart law it gives

$$j(u^*) = -\nabla \times G, \quad G(x; a, d) = \sum_j d_j \ln |x - a_j| + R(x; a, d),$$

where $R(\cdot; a, d)$ is the harmonic boundary correction solving

$$\Delta R = 0 \text{ in } \Omega, \quad R = -\sum_{j=1}^N d_j \ln |x - a_j| \text{ on } \partial\Omega. \quad (7)$$

Working back from $j(u^*)$ to u^* (which is determined up to a global phase) yields the explicit product formula

$$u^*(x; a, d) = \exp(i H(x)) \prod_{j=1}^N \left(\frac{x - a_j}{|x - a_j|} \right)^{d_j}, \quad (8)$$

where $H: \Omega \rightarrow \mathbb{R}$ is the unique zero-mean harmonic function chosen so that u^* satisfies the Neumann boundary condition. Each factor $((x - a_j)/|x - a_j|)^{d_j}$ is the unit complex number pointing from a_j to x , raised to the power d_j ; as x winds once around a_j , this factor rotates by $2\pi d_j$, encoding the correct topological winding.

Note that u^* has genuine singularities at the a_j (it lies in $W^{1,p}(\Omega, S^1)$ only for $p < 2$), which is why it cannot be used directly as initial data for the PDE and must be smoothed out.

1.4 The renormalized energy

The Ginzburg–Landau energy diverges as $\varepsilon \rightarrow 0$ with a logarithmic self-energy $\pi \ln(1/\varepsilon)$ per vortex. After subtracting this divergence one obtains the *renormalized energy*

$$W(a, d) = -\pi \sum_{1 \leq i \neq j \leq N} d_i d_j \ln |a_i - a_j| - \pi \sum_{j=1}^N d_j R(a_j; a, d), \quad (9)$$

so that $E_\varepsilon(u_\varepsilon) - N(\pi \ln(1/\varepsilon) + \gamma) \approx W(a, d)$ as $\varepsilon \rightarrow 0$ for an explicit constant $\gamma > 0$. The first sum captures *vortex–vortex interaction*: same-sign vortices repel ($d_i d_j = +1$ makes the $-\ln$ term large and negative when $a_i \approx a_j$) and opposite-sign vortices attract. The second sum captures *vortex–boundary interaction* through the harmonic correction R . We also define the full approximation

$$W_\varepsilon(a, d) = N\left(\gamma + \pi \ln \frac{1}{\varepsilon}\right) + W(a, d),$$

which approximates E_ε for a wave function made of vortices at a .

1.5 Well-prepared initial conditions

Definition 1.1 (Well-prepared initial conditions). A family $(\psi_\varepsilon^0)_{\varepsilon>0}$ is *well-prepared* if, for some $C > 0$, $0 < \alpha < 1$, and ε small enough:

1. There exist vortex positions $a^0 = (a_j^0)_{j=1, \dots, N} \in \Omega^{*N}$ and degrees $d \in \{\pm 1\}^N$ such that

$$\left\| J\psi_\varepsilon^0 - \pi \sum_{j=1}^N d_j \delta_{a_j^0} \right\|_{\dot{W}^{-1,1}} \leq C\varepsilon^\alpha$$

and the vortices are mutually well-separated.

2. The energy is near-optimal:

$$E_\varepsilon(\psi_\varepsilon^0) \leq W_\varepsilon(a^0, d) + C\varepsilon^{1/2}.$$

The key result of Jerrard–Spirn [**JerrardSpirnRefined2008**] is that well-preparedness is *preserved* along the GP flow.

Theorem 1.2 (Jerrard–Spirn). *Let ψ_ε solve (??) with well-prepared initial conditions and initial vortices at a^0 with degrees d . Then for all $0 \leq t \leq \tau_{\varepsilon, a^0}$,*

$$\left\| J\psi_\varepsilon(t) - \pi \sum_j d_j \delta_{a_j(t)} \right\|_{\dot{W}^{-1,1}} \lesssim \varepsilon^\beta, \quad (10)$$

$$\left\| j(\psi_\varepsilon(t)) - j(u^*(a(t), d)) \right\|_{L^{4/3}} \lesssim \varepsilon^\gamma, \quad (11)$$

where $a(t)$ solves (??) and $0 < \beta, \gamma \leq 1/2$.

Theorem ?? is the theoretical foundation of the method: it says the true GP solution stays close to the canonical harmonic map with vortices at the ODE positions, *for small but finite ε* , not just in the limit.

2 Method

2.1 Motivation: why not solve the PDE directly?

Standard approaches for (??) use time-splitting together with a uniform spatial discretisation (Fourier pseudo-spectral or finite elements). These methods are perfectly adequate for smooth solutions, but in the regime of small ε they face a fundamental obstacle: resolving a vortex core of size $\sim \varepsilon$ requires a mesh size $h \ll \varepsilon$ and, due to CFL constraints, a time step $\delta t \ll \varepsilon^2$. The paper reports that a reference simulation for $\varepsilon = 0.01$ took approximately one week on a small cluster. The method presented here removes ε from the discretisation constraints entirely.

2.2 Constructing well-prepared initial conditions

Starting from the canonical harmonic map u^* , one builds a valid initial condition for (??) by multiplying in a radial cutoff that smoothly kills the singularities at the vortex cores:

$$\psi_\varepsilon^*(x; a^0, d) = u^*(x; a^0, d) \prod_{j=1}^N f_\varepsilon(|x - a_j^0|), \quad (12)$$

where $f_\varepsilon: [0, \infty) \rightarrow [0, 1]$ is the unique radially symmetric vortex core profile solving the one-dimensional boundary value problem

$$\frac{1}{r}(rf'_\varepsilon)' - \frac{1}{r^2}f_\varepsilon + \frac{1}{\varepsilon^2}(1 - f_\varepsilon^2)f_\varepsilon = 0, \quad f_\varepsilon(0) = 0, \quad f_\varepsilon(r_0) = 1. \quad (13)$$

By a scaling argument f_ε depends only on the ratio r/ε , so the 1D BVP (??) is solved *once* in a preprocessing step and stored as an interpolator for the entire simulation.

The role of each factor is clear: far from the vortices ($|x - a_j| \gg \varepsilon$) one has $f_\varepsilon \approx 1$, so $\psi_\varepsilon^* \approx u^*$; near vortex j the factor $f_\varepsilon(|x - a_j|)$ smoothly brings the density $|\psi_\varepsilon^*|^2$ to zero on the scale ε , regularising the singularity of u^* .

By Lemma 1 of [JerrardSpirnRefined2008], the function $\psi_\varepsilon^*(a^0, d)$ constructed in (??) is well-prepared in the sense of the definition above.

2.3 Solving the Laplace problem via harmonic polynomials

At each time step the forcing term $F(a)$ of the ODE (??) requires evaluating $\nabla_{a_j} W$, which in turn requires solving the Laplace problem (??) and evaluating ∇R at the vortex positions. On the unit disk $\Omega = B_1(0)$ this is done spectrally via the following three steps.

Define the family of harmonic polynomials $h_1 \equiv 1$, $h_{2m}(x, y) = \operatorname{Re}(x + iy)^m$, $h_{2m+1}(x, y) = \operatorname{Im}(x + iy)^m$ for $m \geq 1$. On the boundary $\partial\Omega$ these restrict to the standard Fourier basis.

1. **Boundary data.** Evaluate

$$g_a(e^{i\theta}) = - \sum_{j=1}^N d_j \ln |(\cos \theta, \sin \theta) - a_j|$$

on a fine angular grid and compute its Fourier coefficients $\hat{g}_a(k)$ for $|k| \leq n$ via a single FFT.

2. **Harmonic extension.** Set

$$R_n(re^{i\theta}) = \sum_{k=-n}^n r^{|k|} \hat{g}_a(k) e^{ik\theta}.$$

Each term $r^{|k|} e^{ik\theta}$ is harmonic, so R_n is harmonic by linearity and matches $P_n g_a$ on $\partial\Omega$. No linear system needs to be solved.

3. **Gradient evaluation.** Differentiate R_n term-by-term analytically and evaluate at each vortex position a_j .

Since g_a is smooth on $\partial\Omega$ as long as the vortices stay away from the boundary, the Fourier coefficients decay faster than any polynomial in k , yielding *spectral accuracy* in n .

2.4 The full algorithm

The method proceeds in three stages, summarised in Figure 3 of [CorsoKemlinMelcherStamm2024].

Preprocessing (done once):

1. Solve the 1D BVP (??) for f_ε and store the result.
2. Compute the initial phase H by solving a Laplace equation via harmonic polynomials.
3. Build the initial wave function $\psi_\varepsilon^0 = \psi_\varepsilon^*(a^0, d)$ via (??).

Time evolution (cheap, seconds on a laptop):

4. Integrate the Hamiltonian ODE (??) using RK4 with time step $\delta t = 10^{-3}$, evaluating $F(a(t))$ at each stage via the harmonic polynomial method above.

Reconstruction (at any desired output time t):

5. Build the approximate wave function

$$\psi_\varepsilon^*(t) = u^*(\cdot; a(t), d) \prod_{j=1}^N f_\varepsilon(|\cdot - a_j(t)|), \quad (14)$$

where $u^*(a(t), d)$ requires solving one further Laplace equation for the updated phase H .

The key observation is that ε enters *only* through step 1. The ODE integration and the reconstruction are completely independent of ε . In contrast, standard PDE solvers require $h \ll \varepsilon$ and $\delta t \ll \varepsilon^2$, so their cost diverges as $\varepsilon \rightarrow 0$. Here, the cost is controlled entirely by the inter-vortex distance ρ_T , which is an $O(1)$ quantity.

The method has two limitations worth stating explicitly: it is valid only up to the first vortex collision (where the ODE blows up), and it cannot capture nonlinear phenomena such as radiation or sound waves triggered by near-collisions.

3 Error Analysis

3.1 Sources of error

The total error between the true GP solution $\psi_\varepsilon(t)$ and the reconstructed approximation $\psi_\varepsilon^*(t)$ has three independent sources, visible in the triangle inequality

$$\|j(\psi_\varepsilon(t)) - j(\psi_\varepsilon^*(t))\|_{L^{4/3}} \leq \underbrace{\|j(\psi_\varepsilon(t)) - j(u^*(a(t), d))\|_{L^{4/3}}}_{\text{GP limit error, } \lesssim \varepsilon^\gamma} + \underbrace{\|j(u^*(a(t), d)) - j(u^*(b(t), d))\|_{L^{4/3}}}_{\text{ODE trajectory error}} + \underbrace{\|j(u^*(b(t), d)) - j(u^*(b(t), d))\|_{L^{4/3}}}_{\text{recons}}$$

where $a(t)$ is the exact Hamiltonian trajectory and $b(t)$ is the numerically computed one. The first term is bounded by Theorem ???. The third term is bounded by $\varepsilon^{1/2}$ using the same

argument as Proposition 1. The middle term requires bounding $|a(t) - b(t)|$, which is the content of Lemma ?? below.

The trajectory error $|a(t) - b(t)|$ itself arises from two sources:

- **PDE discretisation error:** the Laplace problem is solved only approximately with n harmonic polynomials, so $F^n \neq F$.
- **Time discretisation error:** RK4 integrates $\dot{b} = F^n(b)$ with finite time step δt , introducing a global error $O(\delta t^4)$.

3.2 Estimating the trajectory error

We first bound the error on the continuous dynamics (exact time, approximate forcing).

Lemma 3.1 (Continuous dynamical error). *Let $a^0, b^0 \in \Omega^{*N}$, let $a(t)$ solve (??) exactly with forcing F , and let $b(t)$ solve $\dot{b} = F^n(b)$ with the same initial data and degrees. Define*

$$\rho(a) := \min_{j \neq \ell} \{ \text{dist}(a_j, \partial\Omega), |a_j - a_\ell| \}.$$

Suppose $\rho_T := \inf_{t \in [0, T]} \rho(a(t)) > 0$. Then

$$|a(t) - b(t)| \leq C \left(|a^0 - b^0| + \frac{\sup_{s \in [0, T]} \|P_n^\perp g_a(s)\|_{L^2(\partial\Omega)}}{\rho_T^{3/2}} \right) \exp\left(\frac{Ct}{\rho_T^{5/2}}\right), \quad (15)$$

where $g_a(x) = -\sum_j d_j \ln |x - a_j|$.

The term $\|P_n^\perp g_a\|_{L^2(\partial\Omega)}$ measures how well the boundary data is approximated by n Fourier modes. By Lemma 4 of [CorsoKemlinMelcherStamm2024] (spectral estimates for log functions), if Ω is the unit disk then

$$\|P_n^\perp g_a\|_{L^2(\partial\Omega)} \lesssim_\ell \frac{n^{-\ell + \frac{1}{2}}}{\rho_T^{\frac{1}{2}}} \quad \text{for any } \ell \in \mathbb{N},$$

which decays faster than any polynomial in n (spectral convergence).

The time discretisation error is handled separately.

Lemma 3.2 (Time discretisation error). *Let $b(t)$ be the exact solution to $\dot{b} = F^n(b)$ and let $(b_{\delta t}^k)_{k \delta t \leq T}$ be its RK4 approximation with time step δt :*

$$\begin{aligned} b_{\delta t}^0 &= b^0, \\ b_{\delta t}^k &= b_{\delta t}^{k-1} + \delta t \Phi_{\delta t, n}(b_{\delta t}^{k-1}), \end{aligned}$$

where $\Phi_{\delta t, n}$ is the RK4 increment function for F^n . Assume

$$0 < \rho_T = \min\left(\inf_{0 \leq t \leq T} \rho(b(t)), \min_{k \delta t \leq T} \rho(b_{\delta t}^k)\right).$$

Then there exist constants $C_{\rho_T}, \Lambda_{\rho_T} > 0$, depending on ρ_T but not on n , b , or δt , such that

$$\max_{k\delta t \leq T} |b(k\delta t) - b_{\delta t}^k| \leq \delta t^4 \frac{C_{\rho_T}}{\Lambda_{\rho_T}} (e^{\Lambda_{\rho_T} T} - 1). \quad (16)$$

Combining Lemmas ?? and ?? via the triangle inequality gives the main error bound on the ODE trajectory.

Lemma 3.3 (ODE trajectory error). *Under the assumptions of Lemmas ?? and ??, with $a^0 = b^0$, one has*

$$|a(t) - b(t)| \lesssim_{\ell, T} (1 - \rho_T)^n n^{1-\ell} + \delta t^4 \quad \text{for any } \ell \in \mathbb{N} \text{ and } t \leq T, \quad (17)$$

where the implicit constant depends on ℓ , T , and ρ_T but is independent of n and δt .

3.3 Proof sketch of Lemma ??

The proof applies Gronwall's inequality to the error $e(t) = a(t) - b(t)$. Differentiating and using the triangle inequality gives

$$\left| \dot{a}(t) - \dot{b}(t) \right| = |F(a(t)) - F^n(b(t))| \leq E_1(a, b) + E_2(a) + E_3(a, b),$$

where the three terms isolate different sources:

- $E_3(a, b) = \left| \sum_{k \neq j} d_k \left(\frac{b_j - b_k}{|b_j - b_k|^2} - \frac{a_j - a_k}{|a_j - a_k|^2} \right) \right|$: position error in the explicit vortex-vortex term. Bounded via the Lipschitz estimate $|x/|x|^2 - y/|y|^2| \lesssim |x - y| / \min(|x|, |y|)^2$, giving $E_3 \lesssim |a - b| / \rho_T^2$.
- $E_2(a) = |\nabla R_n(a_j; a, d) - \nabla R(a_j; a, d)|$: PDE discretisation error at the correct positions. The difference $R_n - R$ solves a Laplace equation with boundary data $P_n^\perp g_a$; applying the interior gradient estimate (??) gives $E_2 \lesssim \|P_n^\perp g_a\|_{L^2} / \rho_T^{3/2}$, which decays spectrally in n .
- $E_1(a, b) = |\nabla R_n(b_j; b, d) - \nabla R_n(a_j; a, d)|$: position error in the PDE part. Split further by adding and subtracting $\nabla R_n(a_j; b, d)$: the first piece bounds the sensitivity of ∇R_n to vortex positions via the H^s estimate (49), the second bounds the sensitivity to the evaluation point via the interior estimate (43). Both give contributions $\lesssim |a - b| / \rho_T^{5/2}$.

Summing and noting $|Jv| = |v|$ (since J is an isometry):

$$|\dot{e}(t)| \lesssim \frac{|e(t)|}{\rho_T^{5/2}} + \frac{\|P_n^\perp g_{a(t)}\|_{L^2}}{\rho_T^{3/2}}.$$

Gronwall's inequality then gives (??) directly. □

3.4 Main error theorem

Assembling all contributions yields the main result.

Theorem 3.4 (Error on supercurrents). *Let $\psi_\varepsilon(t)$ be the solution of (??) with well-prepared initial conditions and let $\psi_\varepsilon^*(t)$ be the reconstructed approximation. Then for any $t \leq T$ and $\ell \in \mathbb{N}$,*

$$\|j(\psi_\varepsilon^*(t)) - j(\psi_\varepsilon(t))\|_{L^{4/3}} \lesssim_{\ell,T} \varepsilon^\gamma + n^{-\ell+\frac{1}{2}} + \sqrt{(1-\rho_T)^n n^{1-\ell} + \delta t^4}, \quad (18)$$

where $0 < \gamma \leq 1/2$ and the implicit constant is independent of ε , n , and δt .

Two features of (??) are worth highlighting. First, the ε^γ term *decreases* as $\varepsilon \rightarrow 0$: the method is *asymptotically exact*, in contrast to standard PDE solvers whose cost diverges. Second, the discretisation parameters n and δt appear independently of ε , so they can be chosen based on the inter-vortex distance ρ_T alone.

4 Numerical Results

4.1 Vortex trajectories

Figure 2 of [CorsoKemlinMelcherStamm2024] shows trajectories for eight initial configurations computed in the singular limit $\varepsilon \rightarrow 0$ using a RK4 solver with $\delta t = 10^{-3}$ and $n = 64$ harmonic polynomials. The ODE is solved in a few seconds on a personal laptop.

Key observations:

- **Case 1** (two same-sign vortices): the vortices orbit each other on a circle, consistent with the 2D point vortex theory.
- **Case 2** (opposite-sign pair): the two vortices translate together without orbiting, showing attraction.
- **Cases 7–8** (nine vortices): complex collective motion, still computed in seconds.

4.2 Convergence of the ODE solver

Figure 8 of [CorsoKemlinMelcherStamm2024] shows convergence of the trajectory error $|a(T) - b(T)|$ at $T = 1$:

- **Spectral convergence in n** : the log-linear plot is consistent with exponential decay, confirming the spectral accuracy of the harmonic polynomial method.
- **Fourth-order convergence in δt** : confirms the RK4 theory.
- Pre-factors are larger for vortices near each other or near the boundary (small ρ_T), consistent with Lemma ??.

4.3 Accuracy of the wave function approximation

Figures 10–13 of [CorsoKemlinMelcherStamm2024] compare $\psi_\varepsilon^*(t)$ against a finite element reference solution for Case 1 ($\varepsilon \in \{0.01, 0.03, 0.05, 0.07, 0.1\}$, reference computed on a refined mesh with $h = 2 \cdot 10^{-3}$ at a cost of approximately one week of cluster time for $\varepsilon = 0.01$).

- **Figure 10:** vortex trajectories for finite $\varepsilon > 0$ oscillate around the limiting $\varepsilon = 0$ trajectory; the oscillations shrink as $\varepsilon \rightarrow 0$.
- **Figure 11:** all error norms (supercurrents $L^{4/3}$, wave function L^2 , gradient H^1) decrease with ε , as predicted by Theorem ??.
- **Figure 12:** the smoothed reconstruction ψ_ε^* outperforms the raw canonical harmonic map u^* , especially for the supercurrents.
- **Figure 13:** linear regression in $\log \varepsilon$ gives an exponent $\gamma > 1/2$ for the reconstructed solution, better than the theoretical lower bound $\gamma = 1/2$.